DANIEL L. PIMENTEL ALARCÓN

Background Information

Office Address — 330 N Orchard St (WID), 2176, Madison, WI, 53715
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Formal Education

2008 — B.S., Computer Engineering, Instituto Tecnológico Autónomo de México
2008 — B.S., Telematic Engineering, Instituto Tecnológico Autónomo de México
2014 — M.S., Electrical and Computer Engineering, University of Wisconsin-Madison
2014 — M.A., Mathematics, University of Wisconsin-Madison
2016 — Ph.D., Electrical and Computer Engineering, University of Wisconsin-Madison

Current Position

Since 2019 — Assistant Professor, Wisconsin Institute for Discovery, Department of Biostatistics and Medical Informatics, University of Wisconsin-Madison.

Past Positions

2008-2009 — Project Manager, C-SAM, The Pitroda Group Mexico.
2009-2010 — Statistical Analyst, HSBC Mexico.
2016-2017 — Postdoctoral Researcher, Wisconsin Institute for Discovery, University of Wisconsin - Madison.
2017-2019 — Assistant Professor, Department of Computer Science, Georgia State University.

Honors and Awards

2008 — Social Responsibility Award, ITAM.
2009 — Best paper award, Virtual Medicine Symposium, Universidad Panamericana.
2012 — College of Engineering Teaching Award (Gerald Holdridge), University of Wisconsin-Madison.

Publications

Each publication is annotated with my % contribution for each of these categories: Concept Development and Design (CDD), Data acquisition (DA), Analysis (A), Writing (W). For example, 100|100|20|100 means my students or I contributed 100% to CDD, 100% to DA, 20% to A and 100% to W. If a particular category is not relevant to a paper, it is left as NA. (*) indicates a paper that I deem as most noteworthy. Trainee (student or postdoc) in my lab is underscored. h5: h5-index is a measure commonly used to assess a publication's importance in any given field (the higher the better). For reference, the highest h5 in the field is 337 (NeurIPS), and the JMLR (one of the top journals) has an h5-index of 117 with an acceptance rate (AR) of approximately 20%.

Publications Under Review

[UR1] K. Rupashree, S. Baskar, and **D. Pimentel-Alarcón**. ReLU embeddings preserve general clustering structure. 2025.

[UR2] H. Li, M. Nguyen, and **D. Pimentel-Alarcón**. Preventing collapse in contrastive learning with orthonormal prototypes (CLOP). 2025.

Peer-Reviewed Publications

[PR1] X. Rong, R. McAdams, **D. Pimentel-Alarcón**, and C. Adegboro. Machine learning analysis of cardiotocographs (CTGs) for predicting neonatal encephalopathy (NE) among term infants. *Pediatric Academic Societies*, 2025. **(AR 26%)**
50|10|90|75. I am the mentor of the first author of this paper. I designed the machine learning architecture methods to predict NE and suppervised my student's implementation and experimentation. Drs. Adegboro and McAdams provided the data, informed our methods, and kept them grounded on biomedical realities.

[PR2] K. Rupashree, S. Baskar, and **D. Pimentel-Alarcón**. Latent union completion. In *2025 IEEE International Symposium on Information Theory (ISIT)*. IEEE, 2025. **(AR 43%)**
100|100|100|100. I am the senior author of this paper and the mentor of the other authors in this paper. This paper generalizes a method that combines neural networks with the well-known sparse subspace clustering algorithm, to enable it to handle missing data.

[PR3] H. Li, M. Nguyen, and **D. Pimentel-Alarcón**. Deep fusion: Capturing dependencies in contrastive learning via transformer projection heads. In *2025 IEEE International Symposium on Information Theory (ISIT)*. IEEE, 2025. **(AR 43%)**
100|100|100|100. I am the senior author of this paper and the mentor of the first author of this paper. This paper studies the dependencies in data when using Transformers to perform contrastive learning. The goal is to make the process of contrastive learning more analytical and explainable.

[PR4] A. Soni, J. Linderoth, J. Luedtke, and **D. Pimentel-Alarcón**. An integer programming approach to subspace clustering with missing data. *INFORMS Journal on Optimization*, 7(2):83–104, 2025. **(IF 1.6)**
50|5|30|20. I am the senior author of this paper. This is a collaboration with the Linderoth and Luedtke Labs. I was in charge of designing and analyzing all subspace clustering and missing data aspects of the project. The Linderoth and Luedtke Labs were in charge of the mixed integer program aspects of the project, including implementation and testing.

[PR5] **(*)** S. Baskar, K. Rupashree, and **D. Pimentel-Alarcón**. Deep union completion. In *AAAI Conference on Artificial Intelligence*, 2025. **(h5 220; ranked #5)**
100|100|100|100. I am the senior author of this paper and the mentor of the first authors of this paper. This paper sheds new light on the fundamental clustering properties of deep neural networks and develops a novel clustering method that improves the state-of-the-art by a staggering 40%. This paper represents one of the main deliverables of my NSF CAREER project.

[PR6] H. Li, J. Johnson, and **D. Pimentel-Alarcón**. Fusion over the grassmannian for high-rank matrix completion. In *2024 IEEE International Symposium on Information Theory (ISIT)*, pages 1550–1555. IEEE, 2024. **(AR 42%)**
100|100|100|100. I am the senior author of this paper and the mentor of the first author of this paper. Fusion penalties can be used as a regularization for Euclidean clustering. This paper generalizes fusion penalties over the Grassmann manifold in order to perform subspace clustering with missing data.

[PR7] H. Li and **D. Pimentel-Alarcón**. Group-sparse subspace clustering with elastic stars. In *2024 IEEE International Symposium on Information Theory (ISIT)*, pages 1961–1966. IEEE, 2024. **(AR 42%)**
100|100|100|100. I am the senior author of this paper and the mentor of the first author of this paper. Elastic stars can be used as a regularization for Euclidean clustering. This paper explores its use subspace clustering.

[PR8] A. Singh, M. Fromandi, **D. Pimentel-Alarcón**, D. Werling, A. Gasch, and J.-P. Yu. Intrinsic gene expression correlates of the biophysically modeled diffusion mri signal in autism spectrum disorder. *Biological Psychiatry*, 95(10):S137, 2024. **(h5 96)**

10|0|10|10 This paper shows that there is an identifyable correlation between genes and MRI imagery. This is a collaboration with the Yu Lab. I verified the dimensionality reduction analysis showing that the principal components of these two signals is surprisingly correlated.

[PR9] J. B. Rattray, R. J. Lowhorn, R. Walden, P. Márquez-Zacarías, E. Molotkova, G. Perron, C. Solis-Lemus, **D. Pimentel-Alarcón**, and S. P. Brown. Machine learning identification of pseudomonas aeruginosa strains from colony image data. *PLOS Computational Biology*, 19(12):e1011699, 2023. **(h5 94; ranked #3)**
40|10|90|20. This paper was a collaboration with the Brown Lab and the Solís-Lemus Lab. The Brown Lab was in charge of all the wet lab experiments necessary to collect the bacteria colonies images. Together with the Solís-Lemus lab, me and my student designed, implemented, and deployed the machine learning algorithms to classify the strains of these bacteria.

[PR10] **(*)** H. Li and **D. Pimentel-Alarcón**. Visualizing grassmannians via poincare embeddings. In *18th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (VISIGRAPP) -Volume3: IVAPP, pages27-39 ISBN:978-989-758-634-7; ISSN: 2184-4321*, 2023. **(AR 18%; Best student paper award)**
100|100|100|100. I am the senior author of this paper and the mentor of the first author of this paper. The paper presents a novel way to visualize subspaces as points in the Poincare disk. This proved to be a critical tool to analyze how subspaces move in the Grassmann manifold, which has helped understand the behavior of low-dimensionality algorithms such as PCA, matrix completion, and subspace clustering.

[PR11] **(*)** B. Kizaric and **D. Pimentel-Alarcón**. Principal component trees and their persistent homology. In *AAAI Conference on Artificial Intelligence*, 2023. **(h5 220; ranked #5)**
100|100|100|100. I am the senior author of this paper and the mentor of the first author of this paper. This paper presents a nonparametric model selection criteria to identify the best union of subspaces that best fits an arbitrary dataset. As such, this paper represents one of the main deliverables of my NSF CAREER project.

[PR12] Y. Sun, T. M. Maeda, C. Solís-Lemus, **D. Pimentel-Alarcón**, and Z. Buřivalová. Classification of animal sounds in a hyperdiverse rainforest using convolutional neural networks with data augmentation. *Ecological Indicators*, 145:109621, 2022. **(h5 116)**
90|0|90|30. I am the corresponding author of this paper and co-mentor of the first author of this paper. I was in charge of designing a machine learning framework capable of classifying animal sounds. This was a difficult task that standard methods could not handle because of class imbalance and extremely low-sample regime. This is a collaborator with the Buřivalová Lab, who provided the data, and the Solís-Lemus Lab, in charge of testing.

[PR13] K. Srivastava and **D. Pimentel-Alarcón**. A perturbation bound on the subspace estimator from canonical projections. In *2022 IEEE International Symposium on Information Theory (ISIT)*, pages 1617–1622. IEEE, 2022. **(AR 44%)**
100|100|100|100. I am the senior author of this paper and the mentor of the first author of this paper. This paper bounds the error induced by noise when trying to learn a subspace from incomplete data. This is a critical step towards provable low-rank, high-rank, and algebraic matrix completion.

[PR14] U. Mahmood and **D. Pimentel-Alarcón**. Fusion subspace clustering for incomplete data. In *2022 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2022. **(h5 46)**
100|100|100|100. I am the senior author of this paper and the mentor of the first author of this paper. This paper was the first exploration of fusion penalties on the problem of subspace clustering with missing data.

[PR15] H. Li and **D. Pimentel-Alarcón**. Minimum-length trace reconstruction via integer programming. In *2022 58th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, pages 1–8. IEEE, 2022. **(AR 52%)**
100|100|100|100. I am the senior author of this paper and the mentor of the first author of this paper. This paper gives an integer program approach to the classical problem of the minimum-length trace reconstruction. Our methods improve the state of the art for problems of modest size (i.e., dimensions no larger than ∼1000).

[PR16] B. Kizaric and **D. Pimentel-Alarcón**. Classifying incomplete data with a mixture of subspace experts. In *Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, pages 1344–1352. IEEE, 2022. **(AR 52%)**
100|100|100|100. I am the senior author of this paper and the mentor of the first author of this paper. This paper presents a novel approach to classification with large amounts of missing data that is an extension of the union of subspaces model.

[PR17] H. Lin, M. Li, **D. Pimentel-Alarcón**, and M. L. Malloy. Geometry of the minimum volume confidence sets. In *2022 IEEE International Symposium on Information Theory (ISIT)*, pages 3180–3185. IEEE, 2022. **(AR 44%)**
10|0|20|10. This paper is a collaboration with the Malloy Lab. I was in charge of the geometric analysis.

[PR18] A. Soni, J. Linderoth, J. Luedtke, and **D. Pimentel-Alarcón**. Integer programming approaches to subspace clustering with missing data. *Optimization for Machine Learning, NeurIPS*, 2021. **(h5 337; ranked #1)**
50|5|30|20. I am the senior author of this paper. This is a collaboration with the Linderoth and Luedtke Labs. I was in charge of designing and analyzing all subspace clustering and missing data aspects of the project. The Linderoth and Luedtke Labs were in charge of the mixed integer program aspects of the project, including implementation and testing.

[PR19] A. Pimentel-Alarcón and **D. Pimentel-Alarcón**. Mixed-features vectors and subspace splitting. In *International Conference on Learning Representations*, 2021. **(h5 304; ranked #2)**
100|20|100|100. I am the senior author of this paper with my brother, who assisted me running some experiments. The paper introduces a new dimensionality reduction model where different subsets of features lie in different subspaces. We also introduce a method to identify such subsets.

[PR20] G. Ongie, **D. Pimentel-Alarcón**, L. Balzano, R. Willett, and R. D. Nowak. Tensor methods for nonlinear matrix completion. *SIAM Journal on Mathematics of Data Science*, 3(1):253–279, 2021. **(h5 80)**
20|20|40|20. This paper is an extended version of our previous paper 'Low algebraic dimension matrix completion', of which I am first author.

[PR21] R. Walden, D. Hope, S. Jefferies, and **D. Pimentel-Alarcón**. Satellite characterization via restoration-free imaging: A novel machine learning paradigm for ssa. In *Advanced Maui Optical and Space Surveillance Technologies (AMOS)*, 2020. **(h5 Not Available)**
80|80|80|70. I am the senior author of this paper and the mentor of the first author of this paper. Drs. Jefferies and Hope provided the dataset, I designed the algorithm, and my student implemented and applied it.

[PR22] C. Solís-Lemus, X. Ma, M. Hostetter, S. Kundu, P. Qiu, and **D. Pimentel-Alarcón**. Prediction of functional markers of mass cytometry data via deep learning. In *Statistical Modeling in Biomedical Research*, pages 95–104. Springer, 2020. **(h5 47)**
20|20|20|20. I am the senior author of this paper, which was a collaboration with the Kundu and Solís-Lemus Labs, who were in charge of providing and pre-processing the data. My student and I designed, implemented, and tested the deep learning architecture.

[PR23] M. M. Rahman and **D. Pimentel-Alarcón**. Glimps: A greedy mixed-integer approach for super robust matched subspace detection. In *2019 57th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, pages 360–367. IEEE, 2019. **(AR 53%)**
100|100|100|50. I am the senior author of this paper and the mentor of the first author of this paper. It explores integer programming to detect signals corrupted by large amounts of outliers.

[PR24] **D. Pimentel-Alarcón**, A. Tiwari, D. Hope, and S. Jefferies. Leveraging machine learning for high-resolution restoration of satellite imagery. In *Advanced Maui Optical and Space Surveillance Technologies (AMOS)*, 2018. **(h5 Not Available)**
50|100|100|50. I am the senior and first author of this paper, which uses a deep neural network to restore satellite images corrupted by non-linear atmospheric transformations. Drs. Jefferies and Hope provided the dataset. I designed the algorithm, and my student implemented and applied it.

[PR25] **D. Pimentel-Alarcón**. Selective erasures for high-dimensional robust subspace tracking. In *Compressive Sensing VII: From Diverse Modalities to Big Data Analytics*, volume 10658, page 1065808. International Society for Optics and Photonics, 2018. **(h5 Not Available)**
100|100|100|100. This paper gives a greedy algorithm for subspace tracking when there is a large amount of outliers.

[PR26] **D. Pimentel-Alarcón**. Mixture matrix completion. *Advances in Neural Information Processing Systems (NeurIPS)*, 31, 2018. **(h5 337; ranked #1)**
100|100|100|100. This paper presents a novel model where each entry in a data matrix corresponds to one of several low-dimensional models. It is inspired by metagenomics and recommender systems, and is a generalization of low- and high-rank matrix completion.

[PR27] C. R. Solis-Lemus and **D. Pimentel-Alarcón**. Breaking the limits of subspace inference. In *2018 56th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, pages 654–661. IEEE, 2018. **(AR 53%)**
50|50|50|50. Both authors contributed equally to this paper. It is well-known that learning an r-dimensional subspace requires at least r+1 observations per sample. Surprisingly and counterintuitively, this paper shows that this task is actually feasible with only 2 observations per sample under certain conditions.

[PR28] **D. Pimentel-Alarcón**, G. Ongie, L. Balzano, R. Willett, and R. Nowak. Low algebraic dimension matrix completion. In *2017 55th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, pages 790–797. IEEE, 2017. **(AR 52%)**
20|40|40|30. This paper was a collaboration with my former PhD advisor, and the Willett and Balzano Labs. Low-rank matrix completion assumes that data lies near a linear subspace. This paper assumes that data lies near an algebraic variety. It uses a lifting, and one of my previous fundamental results on learning subspaces from canonical projections. I was in charge of the analysis of the later.

[PR29] **D. Pimentel-Alarcón** and R. Nowak. Random consensus robust pca. In *Artificial Intelligence and Statistics*, pages 344–352. PMLR, 2017. **(h5 100 – Oral presentation, top 5%)**
100|100|900|90. This was a collaboration with my former PhD advisor. This paper introduces a novel method for robust PCA that makes no distributional assumptions on the data, and applies to arbitrary locations in the outliers, as opposed to typical methods that assume outliers are uniformly distributed.

[PR30] **D. Pimentel-Alarcón**, A. Biswas, and C. R. Solís-Lemus. Adversarial principal component analysis. In *2017 IEEE International Symposium on Information Theory (ISIT)*, pages 2363–2367. IEEE, 2017. **(AR 42%)**
100|80|90|80. I am the first and senior author of this paper, which gives a bound on how much the principal components can be perturbed.

[PR31] **D. Pimentel-Alarcón**, L. Balzano, R. Marcia, R. Nowak, and R. Willett. Mixture regression as subspace clustering. In *2017 International Conference on Sampling Theory and Applications (SampTA)*, pages 456–459. IEEE, 2017. **(AR Not Available)**
100|100|90|80. This paper was written under the supervision of my former PhD advisor. It shows that mixture regression is a special case of subspace clustering.

[PR32] **D. Pimentel-Alarcón** and C. R. Solís-Lemus. Crime detection via crowdsourcing. In *Mexican Conference on Pattern Recognition*, pages 313–323. Springer, 2016. **(h5 Not Available)**
50|50|50|50. Both authors contributed equally. This paper shows an interesting, efficient, and provable approach to detect crime in communities.

[PR33] **D. Pimentel-Alarcón** and R. D. Nowak. A converse to low-rank matrix completion. In *2016 IEEE International Symposium on Information Theory (ISIT)*, pages 96–100. IEEE, 2016. **(AR 52%)**
100|100|100|90. This paper was written during my PhD. It develops the foundations for model selection when data is missing.

[PR34] **D. Pimentel-Alarcón** and R. Nowak. The information-theoretic requirements of subspace clustering with missing data. In *International Conference on Machine Learning*, pages 802–810. PMLR, 2016. **(h5 268; ranked #3)**
80|100|80|70. This paper was written during my PhD. It establishes the sampling conditions to recover a union of subspaces when data is missing, answering an important open question that has informed subsequent algorithms and theory in the field.

[PR35] **D. Pimentel-Alarcón**, N. Boston, and R. D. Nowak. A characterization of deterministic sampling patterns for low-rank matrix completion. *IEEE Journal of Selected Topics in Signal Processing*, 10(4):623–636, 2016. **(h5 75)**
90|100|90|80. This paper was written during my PhD. It establishes the sampling conditions to recover a low-dimensional linear subspace from incomplete data. It answers an important open question that has informed subsequent algorithms and theory in the field, and sheds some light on the century-old open problem of the characterization of the determinantal variety.

[PR36] **D. Pimentel-Alarcón**, L. Balzano, and R. Nowak. Necessary and sufficient conditions for sketched subspace clustering. In *2016 54th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, pages 1335–1343. IEEE, 2016. **(AR 52%)**
80|80|80|70. This paper was written during my PhD. It characterizes special conditions for clustering when data is missing.

[PR37] **D. Pimentel-Alarcón**, L. Balzano, R. Marcia, R. Nowak, and R. Willett. Group-sparse subspace clustering with missing data. In *2016 IEEE Statistical Signal Processing Workshop (SSP)*, pages 1–5. IEEE, 2016. **(AR Not Available)**
30|30|40|60. This paper was written during my PhD. It introduced the state-of-the-art algorithm (which still remains competitive 10 years later) for subspace clustering with missing data.

[PR38] **D. Pimentel-Alarcón**. A simpler approach to low-rank tensor canonical polyadic decomposition. In *2016 54th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, pages 474–481. IEEE, 2016. **(AR 44%)**
100|100|100|100. The canonical polyadic decomposition can be seen as a generalization of PCA to data arrays of size larger than 2. This paper derives a deterministic and optimal method for such decomposition when the rank of the tensor is lower than its smallest dimension.

[PR39] **D. Pimentel-Alarcón** and R. D. Nowak. Adaptive strategy for restricted-sampling noisy low-rank matrix completion. In *2015 IEEE 6th International Workshop on Computational Advances in Multi-Sensor*

Adaptive Processing (CAMSAP), pages 429–432. IEEE, 2015. **(AR Not Available)**
90|100|100|90. This paper was written during my PhD. It exploits our characterization of sampling patterns for low-rank matrix completion to derive an optimal algorithm in the case where the observed samples can be selected among a subset.

[PR40] **D. Pimentel-Alarcón**, N. Boston, and R. D. Nowak. Deterministic conditions for subspace identifiability from incomplete sampling. In *2015 IEEE International Symposium on Information Theory (ISIT)*, pages 2191–2195. IEEE, 2015. **(AR 51%)**
80|100|90|40. This paper was perhaps the most critical result of my PhD. It took us nearly 4 years of relentless research. It characterizes when and how a subspace can be reconstructed from its partial observations on subsets of coordinates. This result is a cornerstone to characterize how to recover subspaces, unions of subspaces and algebraic varieties from incomplete data, and to derive model selection criteria when data is missing.

[PR41] **D. Pimentel-Alarcón**, R. Nowak, and L. Balzano. On the sample complexity of subspace clustering with missing data. In *2014 IEEE Workshop on Statistical Signal Processing (SSP)*, pages 280–283. IEEE, 2014. **(AR Not Available)**
10|80|80|30. This was the first paper of my PhD. It explores an EM algorithm to achieve subspace-clustering with missing data, and it finds bounds on the amount of data necessary for this task, which at that point remained an open question that was later answered by our subsequent work.

[PR42] **D. Pimentel-Alarcón**, M. E. Algorri, M. Julian, and A. Germán. Non-rigid image registration based on form transformations. In *Virtual Medicine Symposium*, pages 271–277, 2009. **(h5 Not Available #; Best Paper Award)**
30|100|80|70. This paper was written during my bachelor program. it uses thin-plate splines to register thoraxic MRIs.

[PR43] **D. Pimentel-Alarcón**, M. E. Algorri, M. Julian, and A. Germán. Multimodal image registration with thin-plate splines. In *Mexican Conference on Pattern Recognition*, pages 128–133. Springer, 2009. **(h5 Not Available)**
30|100|80|70. This paper was written during my bachelor program. it uses thin-plate splines to register a variety of medical images.

Invited Research Presentations

2025 — **Keynote** at Mexican Conference in Pattern Recognition, Centro de Investigaciones Matemáticas (CIMAT), Mexico, *Nonnegative Subspace Edges (NoSEs) and an optimal solution to NMF*.
2025 — **Keynote** at National Optimization and Numerical Analysis Workshop (ENOAN), Universidad de Guanajuato, Mexico, *Clustering in closed-form: myth or reality?*.
2025 — Midwest Machine Learning Symposium, University of Chicago, *Fantastic Nonnegative Subspace Edges and Where to Find Them*.
2025 — Center for Signal and Information Processing (CSIP) Seminar at Georgia Institute of Technology, *Nonnegative rank: myths, conjectures, and canonical edges*.
2025 — Statistics Seminar at Colorado State University, *Universal clustering in closed form: myth or reality?*.
2025 — Computer Science Seminar at Georgia State University, *Nonnegative matrix factorization: rank unicorns, conjectures, and canonical edges*.
2025 — Allerton Conference on Communication, Control, and Computing, University of Illinois Urbana-Champaign, *Nonnegative Subspace Edges and Deterministic Nonnegative Matrix Completion*.
2024 — Electrical and Computer Engineering Seminar at Portland State University, *Closed-form clustering and the mysteries around it*.

2024 — Biostatistics and Medical Informatics Seminar at UW-Madison, *Unsupervised Learning from Messy Data*.
2023 — WID Seminar series at UW-Madison, *Models and microbes and macs - Oh my*.
2023 — American Family Funding Initiative Networking Meeting at UW-Madison, *Heterogeneous Matrix Completion*.
2023 — Celebrating Latinx voices in STEM at UW-Madison, *The mathematics of the waffle cone*.
2022 — Applied Algebra Seminar at UW-Madison, *Intersecting Schubert Varieties for Subspace Clustering*.
2021 — El Zoominario, *Machine Learning: Inside the Black Box*.
2020 — SIAM Conference on Mathematics of Data Science: Advances in Subspace Learning and Clustering, *Fusion Subspace Clustering for Missing Data*.
2020 — Systems Information Learning Optimization Seminar, *Unsolved Subspace Mysteries*.
2019 — The Center for Robust Decision-making on Climate and Energy Policy, UChicago, *Robust Learning from Incomplete and Messy Data*.
2018 — NSF TRIPODS: Nonconvex Formulations & Algorithms in Data Science, *Fusion Subspace Clustering: Full & Missing Data*.
2018 — Bioinformatics User Group Seminar in Georgia Tech (BUGS), *Decision Trees and Random Forests*.
2017 — Center for Signal and Information Processing Seminar at Georgia Tech, *Matrix Completion goes Rogue (Nonlinear)!*
2017 — Georgia Statistics Day, *Machine Learning: who is teaching?*
2017 — SIAM Conference on Optimization, *Mixture Matrix Completion*.
2017 — ECE Seminar at RICE, *Learning Subspaces by Pieces*.
2016 — Applied Algebra Days at UW-Madison, *Learning Subspaces by Pieces*.
2016 — Systems Information Learning Optimization Seminar, *Random Consensus Robust PCA*
2015 — Applied Algebra Seminar at UC-Berkeley, *On Subspaces and Missing Data*.
2015 — Annual Workshop on Data Sciences at Tennessee State University, *On the Difficulties of Subspace Clustering with Missing Data*.
2014 — Applied Algebra Seminar at UW-Madison, *Identifying Subspaces from Canonical Projections*.
2014 — Systems Information Learning Optimization Seminar, *Deterministic Conditions for Subspace Identifiability from Incomplete Sampling*.
2009 — Virtual Medicine Symposium, *Non-Rigid Image Registration Based on Form Transformations*.

Peer-Reviewed Presentations

2019 — GLIMPS: A greedy mixed-integer approach for super robust matched subspace detection, *Allerton Conference on Communication, Control, and Computing*.
2018 — Leveraging Machine Learning for High-Resolution Restoration of Satellite Imagery, *Advanced Maui Optical and Space Surveillance Technologies (AMOS)*.
2018 — Selective Erasures for High-Dimensional Robust Subspace Tracking, *SPIE Defense + Commercial Sensing*.
2017 — Adversarial Principal Component Analysis, *International Symposium on Information Theory (ISIT)*.
2017 — Random Consensus Robust PCA*Artificial Intelligence and Statistics (AISTATS)*.
2016 — A Simpler Approach to Low-Rank Tensor Canonical Polyadic Decomposition, *Allerton Conference on Communication, Control, and Computing*.
2016 — The Information-Theoretic Requirements of Subspace Clustering with Missing Data, *International Conference on Machine Learning (ICML)*.
2016 — A Converse to Low-Rank Matrix Completion, *International Symposium on Information Theory*.
2016 — Necessary and Sufficient Conditions for Sketched Subspace Clustering, *Allerton Conference on Communication, Control, and Computing*.
2016 — Crime Detection via Crowdsourcing, *Mexican Workshop in Pattern Recognition*.

- 2015 — A Characterization of Deterministic Sampling Patterns for Low-Rank Matrix Completion, *Allerton Conference on Communication, Control, and Computing*.
- 2015 — Deterministic Conditions for Subspace Identifiability from Incomplete Sampling, *International Symposium on Information Theory*
- 2009 — Multimodal Image Registration with Thin-Plate Splines, *Mexican Workshop in Pattern Recognition*.

Research Support

Current

- NSF CAREER: Learning and selecting low-dimensional models from incomplete data
  Role: PI (10% effort)
  National Science Foundation, $600k, 2023-2027

- NIH-R01: Using redox balance to guide surgical and therapeutic decisions for cartilage disease
  PI: Corinne Henak, Role: Co-I (10% effort)
  National Institutes of Health, $1,487,445, 2024-2028

- Nonnegative matrix factorization through nonnegative canonical edges
  Role: PI (1% effort)
  Fall Competition, UW-Madison, $50k, 2024-2025.

Pending

- Nonnegative matrix factorization through cone collapsing and expanding
  Role: PI (10% effort)
  National Science Foundation, $600k, 2025-2028.
- MIRA: Discovering biomedical processes through nonnegative subspace clustering
  Role: PI (25% effort)
  National Institutes of Health, $1000,000, 2025-2030.
- R21: From contrastive learning to transformers: Unlocking the power of deep learning to predict response to interventions from multi-omics data.
  Role: PI (10% effort)
  National Institutes of Health, $400,000, 2025-2030.
- Imaging gene expression in the brain with topological diffusion mapping MRI
  PI: John-Paul Yu, Role: Co-PI (10% effort)
  Research Forward, UW-Madison, $500k, 2025-2027
- Using AI to Transform Biodiversity and Forest Conservation
  PI: Holly Gibbs. Role: Co-PI (10% effort)
  Research Forward, UW-Madison, $500k, 2025-2027

Past

- Towards rigidity through matrix completion and the Plücker embedding
  Role: PI (1% effort)
  Fall Competition, UW-Madison, $50k, 2023-2024.
- Optimal Features for Heterogeneous Matrix Completion
  Role: PI (10% effort)
  American Family, $100k, 2022-2023.
- Integer Programming for Mixture Matrix Completion
  PI: Jeff Linderoth, Role: Co-PI (3% effort)
  American Family, $150k, 2020-2022.

- Analysis of Fetal Heart Rate Tracings to Determine Signs of Neonatal
  PI: Claudette Adegboro, Role: Collaborator
  UnityPoint Health Meriter Hospital, $10k, 2022.

Educational Activities & Presentations

Classroom Teaching (Graduate & Undergraduate)						
Years	Course Title	Credits	Students	Contact Hours	Format	Evals
2020	BMI 567 (Med Im Proc)	3	∼30	4.5h/week	Lecture	3.8/5
2020	CS 760 (Mach Learn)	3	∼120	4.5h/week	Lecture	4.76/7
2022	CS 760 (Mach Learn)	3	∼120	4.5h/week	Lecture	5.51/7
2022	PACE+	3	8	3h/week	Facilitator	N/A
2024	CS 760 (Mach Learn)	3	∼120	4.5h/week	Lecture	4.77/7

Guest Lectures (Graduate & Undergraduate)					
Years	Course Title	Credits	Students	Contact Hours	Format
2021	ENVIR ST 400 (Bioacoustics for Conservation)	3	∼25	1h15	Lecture
2020	ECE 697 (Mach Learn & Sign Proc Capstone)	3	∼30	50 mins	Lecture
2021	ECE 697 (Mach Learn & Sign Proc Capstone)	3	∼30	50 mins	Lecture
2022	ECE 697 (Mach Learn & Sign Proc Capstone)	3	∼30	50 mins	Lecture
2023	ECE 697 (Mach Learn & Sign Proc Capstone)	3	∼30	50 mins	Lecture

Graduate Student Mentees			
Years	Mentee Name	Degree Program	Current Position
2020-	Huanran Li	PhD (ECE)	In progress
2021-2024	Jeremy Johnson	PhD (Math)	Graduated
2022-2023	Ben Kizaric	MS (ECE MLSP)	Graduated
2023-	Karan Rupashree	PhD ECE	In progress
2023-	Siddharth Baskar	PhD ECE	In progress
2023-	Xiaoxu Rong	PhD (EdPsy)	In progress
2024-	Michael Fromandi	BDS	In progress

Undergraduate Student Mentees			
Years	Mentee Name	Degree Program	Current Position
2019-2021	Yuren Sun	CS	Stanford MS
2022-2023	Zihao Li	CS	University of Southern California MS
2022-	Manh Nguyen	CS	In progress

Service Activities

2023-2024 — Shakhashiri Award Committee.

Since 2023 — BDS PhD Admissions Committee.
2022-2023 — Shakhashiri Award Committee.
2021-2022 — WID Mural Committee.
2021 — BMI Machine Learning Faculty Search Committee.
2020 — BMI/WID Machine Learning Faculty Search Committee.
2017 — Program committee, Workshop on Solar & Stellar Astronomy BIg Data (SABID).
2017 — Workflow committee, International Conference on Artificial Intelligence and Statistics (AISTATS).
2015-2016 — Systems Information Learning Optimization Seminar, UW-Madison.

Professional Services

2025 — NSF CISE CCF CIF Pannel.
2022 — NSF CISE CCF CIF Pannel.
Since 2014 — Reviewer for IEEE Transactions on Signal Processing, Neural Information Processing Systems (NIPS), IEEE Signal Processing Letters, Transactions on Pattern Analysis and Machine Intelligence (TPAMI), International Conference in Machine Learning (ICML), International Conference on Artificial Intelligence and Statistics (AISTATS), Journal for Machine Learning Research (JMLR).

Community Outreach/Service

2025 — Co-organizer of *Celebrating Latinx voices in STEM Symposium*, Wisconsin Institute for Discovery.
2024 — Hispanic Scientist Panel at Arcadia Public Library, *The black box of Machine Learning*.
2024 — Science and Improv Comedy at the Wisconsin Science Festival, *Machine Learning: a child's game*.
2024 — Co-organizer of *Celebrating Latinx voices in STEM Symposium*, Wisconsin Institute for Discovery.
2023 — Co-creator of *Celebrating Latinx voices in STEM Symposium*, Wisconsin Institute for Discovery.
2022 — Latinx in STEM, WI Public Television, Wisconsin Latinx History Collective (WLHC).
Since 2021 — Co-creator of *El Zoominario K-12*: short talks by LatinX scientists for at young students.
Since 2020 — Co-creator of *El Zoominario*: scientific contributions by the LatinX Community.
2005-2008 — Math high-school teacher for adults, ITAM, Mexico.